

(Sprint 4)Sprint Planning Meeting/Technical discussion

Date	Time	Location
9 Sept 2026	1pm	Wechat(On line)

 Participants

- Yijia Chen
- Jiajun Huang
- Zeming Xia
- Tong Zhao

 Agenda Items

Agenda Item	Things to discuss
feedback on existing figma	
Plan sprint action items	

 Discussion topics

Topic	Notes

Technical discussion

1. UV Data & Location Repository

Sprint 2 feature branches have been merged. `RealUVRepo` is integrated to replace mock UV data. The team needs to fix dependency injection for `FusedCurrentLocationProvider` to pass Context and retrieve real location. Mock data is kept only for the unfinished light sensor module.

2. Sun Exposure Countdown Timer

The timer estimates exposure dose; a manual lux slider is used before the light sensor is ready. New rule: individual dose limit = 48% of MED, with a 2.4 SED hard cap for Fitzpatrick Type V & VI to avoid extreme long values. The field is renamed `minimumErythemaDoseSed`. Switching skin type updates the limit without clearing accumulated UV dose. The +10 min quick button is hidden when UV index = 0. The result is marked as a protective-action estimate instead of guaranteed safe time.

3. SPF & Sunscreen Logic

SPF stays visible and selectable on UI but is excluded from countdown multiplication. Backend adds status enums: `BELOW_RECOMMENDED` and `RECOMMENDED`. A separate sunscreen reapplication reminder (default 2 hours) will be built, resettable after swimming or wiping. This follows WHO and SunSmart guidance, as labelled SPF overestimates real protection. SPF may affect dose calculation later only with reliable sensor data.

4. Light Sensor Module

Ambient light sensor is pending. Live lux values will replace the manual slider and auto-update exposure estimates once implemented.

Plan sprint action items	1. Resolve Context injection for <code>FusedCurrentLocationProvider</code> inside <code>RealUVRepo</code> to load live location and UV API data. 2. Implement the updated timer algorithm (48% MED with 2.4 SED cap) and hide the +10 minute button when UV index equals zero. 3. Build the ambient light sensor and connect real lux readings to exposure estimation. 4. Develop SPF status display and standalone sunscreen reapplication reminder UI.
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Action items

- ☐ Zeming Xia : Resolve context injection issue for `FusedCurrentLocationProvider` and implement `RealUvRepo` to feed live API UV data to frontend
- ☐ Team: Finalise and submit the revised countdown timer calculation logic with SED upper limits
- ☐ Team : Remove the +10 min timer button for zero UV index scenarios
- ☐ Team: Implement light sensor integration and connect sensor readings to the brightness slider
- ☐ Team: Plan future sunscreen reapplication timer and SPF warning UI